Large-Signal Analysis of the *CMOS* Differential Pair

Undergrad *CMOS* Course

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Why do we need to study Differential Amplifiers?

The elegant concept of "*differential*" signals and amplifiers was invented in the 1940's and first utilized in vacuum-tube circuits. Since then, differential circuits have found increasingly wider usage in microelectronics and serve as a robust, high-performance design paradigm in many of today's systems.

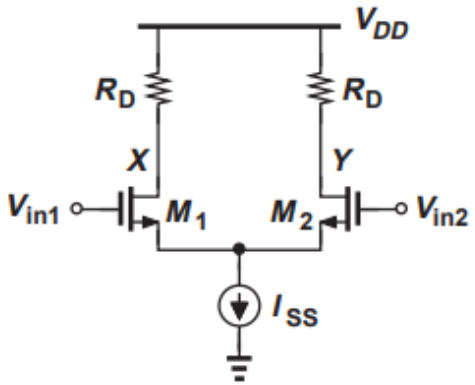


Figure 1: *MOS* Differential Amplifier. [1]

In Fig. (1), we can see the *MOS* Differential Amplifier that we'll be studying in this presentation.

Introductory Example

We know that *OPAMP's* have 2 inputs, a point of contrast to the amplifiers studied in previous classes of this course. In order to further understand the need for differential circuits, let us first consider an introductory example.

Example 1:

Let's suppose that a fellow classmate constructs the circuit shown in Fig. 2 to amplify the signal produced by a microphone. Unfortunately, upon applying the result to a speaker, the classmate observes that the amplifier output contains a strong *"humming"* noise, this being a steady low-frequency component. Explain what happens.

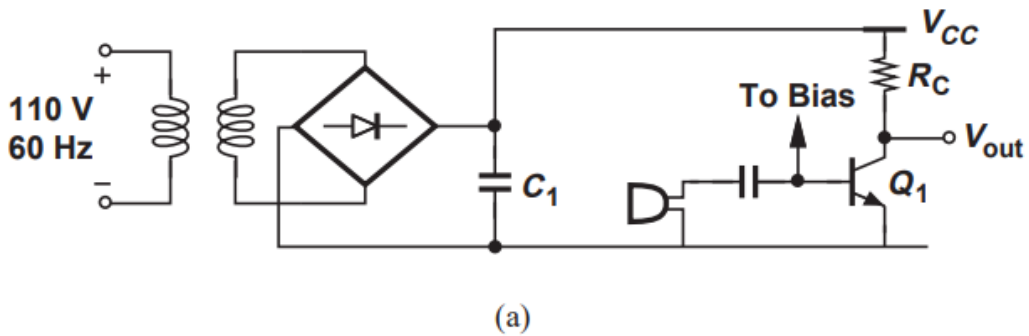


Figure 2: Microphone amplifier designed by a fellow classmate. [2]

Introductory Example: Understanding the Problem

The current drawn from the rectified output creates a ripple waveform at twice the AC line frequency (50 or 60 $[Hz]$) as shown in Fig. (3.b)

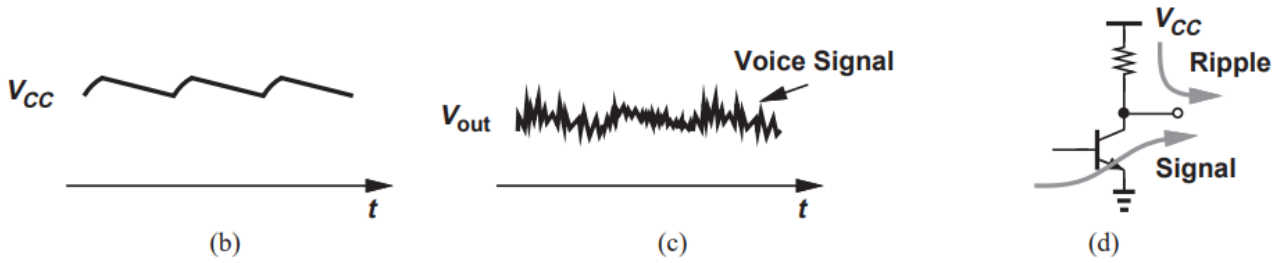


Figure 3: Microphone amplifier designed by a fellow classmate. [2]

Examining the output of the common-emitter stage, we can identify two components:

- 1) The amplified version of the microphone signal.
- 2) The ripple waveform present on V_{CC}.

For the latter, we can write:

$$V_{out} = V_{CC} - I_C \cdot R_C \tag{1}$$

Introductory Example: Understanding the Problem

Noting that V_{out} simply "tracks" V_{CC} in Eq. (1) and hence contains the ripple in its entirety. Note also that the "hum" originates from the ripple.

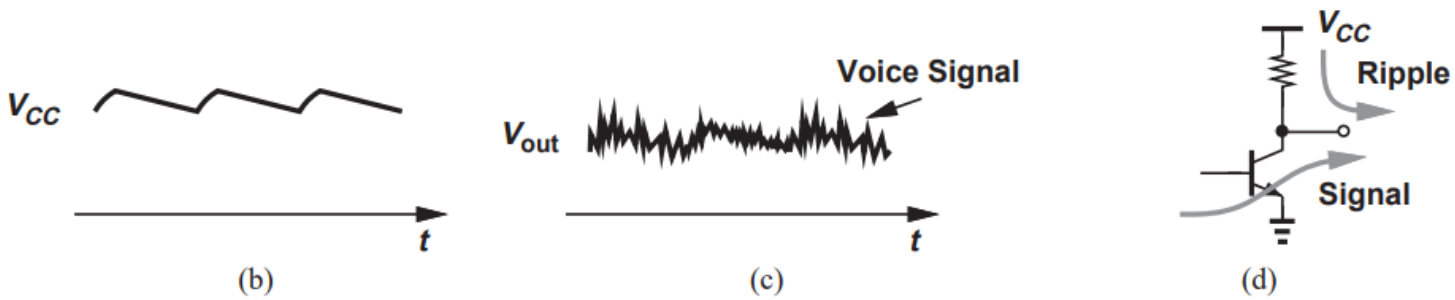


Figure 4: Microphone amplifier designed by a fellow classmate. [2]

Fig. (4.c) depicts the overall output in the presence of both the signal and the ripple. Illustrated in Fig. (4.d), this phenomenon is summarized as the "supply noise goes to the output with a gain of unity."

Note that a **MOS implementation would suffer from the same problem**. Therefore, we wonder: How should we suppress the hum in the above example?

Introductory Example: How to fix it?

Solution 1:

We can try increasing C_1, thus lowering the ripple amplitude, but the required capacitor value may become prohibitively large if many circuits draw current from the rectifier. $\longrightarrow$ NOT RELIABLE

Solution 2:

We can modify the amplifier topology such that the output is insensitive to V_{CC}.

- Eq. (1) implies that a change in V_{CC} directly appears in V_{out}, fundamentally because both V_{out} and V_{CC} are measured with respect to ground and differ by a factor of $I_C R_C$.
- But what if V_{out} is not "referenced" to ground? More specifically, what if V_{out} is measured with respect to another point that itself experiences the supply ripple to the same extent? It is thus possible to eliminate the ripple from the "net" output.

COULD WORK!

Introductory Example: The Solution

While rather abstract, the above conjecture can be readily implemented as shown in Fig. (5.a) which illustrates the core concept of the solution to this problem.

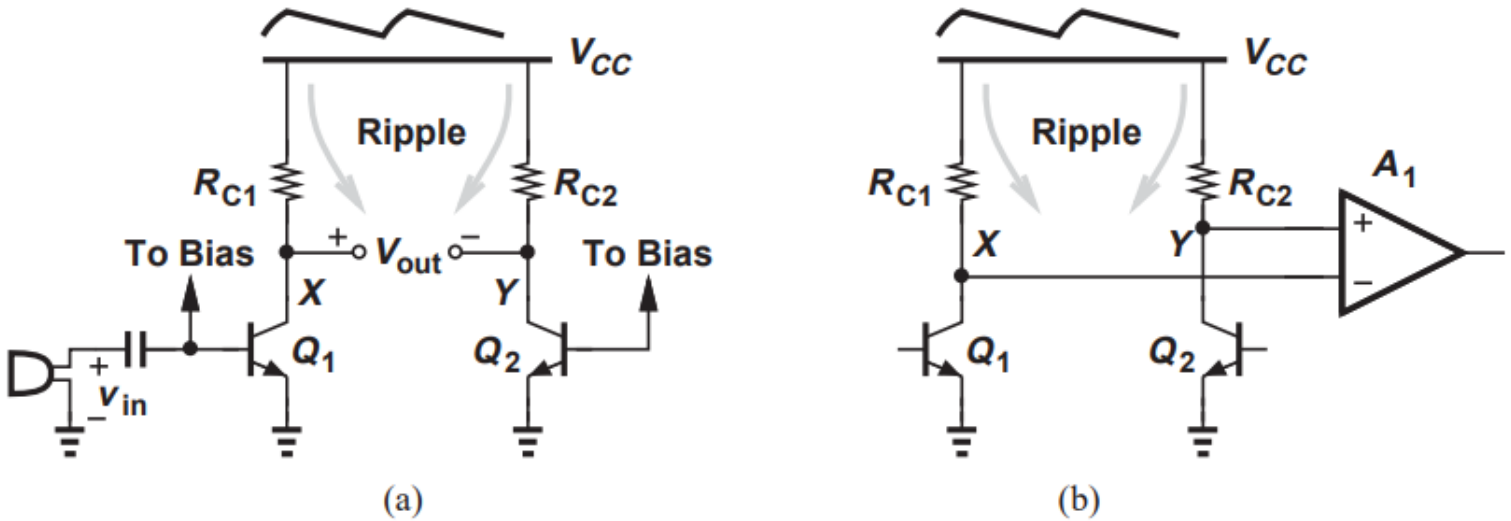


Figure 5: Solution to the *humming* problem. [3]

The *CE* (or *CS* for MOS amplifiers) stage is duplicated on the right, and the output is now measured between nodes X and Y rather than from X to ground.

Introductory Example: The Solution

One may wonder then: *what happens if V_{CC} contains ripple?* Well, note that both V_X and V_Y rise and fall by the same amount and hence the difference between V_X and V_Y remains free from the ripple.

In fact, denoting the ripple by v_r, we express the small-signal voltages at these nodes as:

$$\begin{cases} v_X = A_v \cdot v_{in} + v_r \\ v_Y = v_r \end{cases} \tag{2}$$

That is:

$$v_X - v_Y = A_v \cdot v_{in} \tag{3}$$

Note that Q_2 carries no signal, simply serving as a constant current source. The above development serves as the foundation for **differential amplifiers**: the symmetric CE stages provide two output nodes whose voltage difference remains free from the supply ripple.

Differential Signals: Analysis of the Solution

Let us return to the circuit of Fig. (5.a) and recall that the duplicate stage consisting of Q_2 and R_{C2} remains "idle", thereby "wasting" current.

We may therefore wonder if this stage can provide signal amplification in addition to establishing a reference point for V_{out}. In our first attempt, we directly apply the input signal to the base of Q_2 as shown in Fig. (6.a).

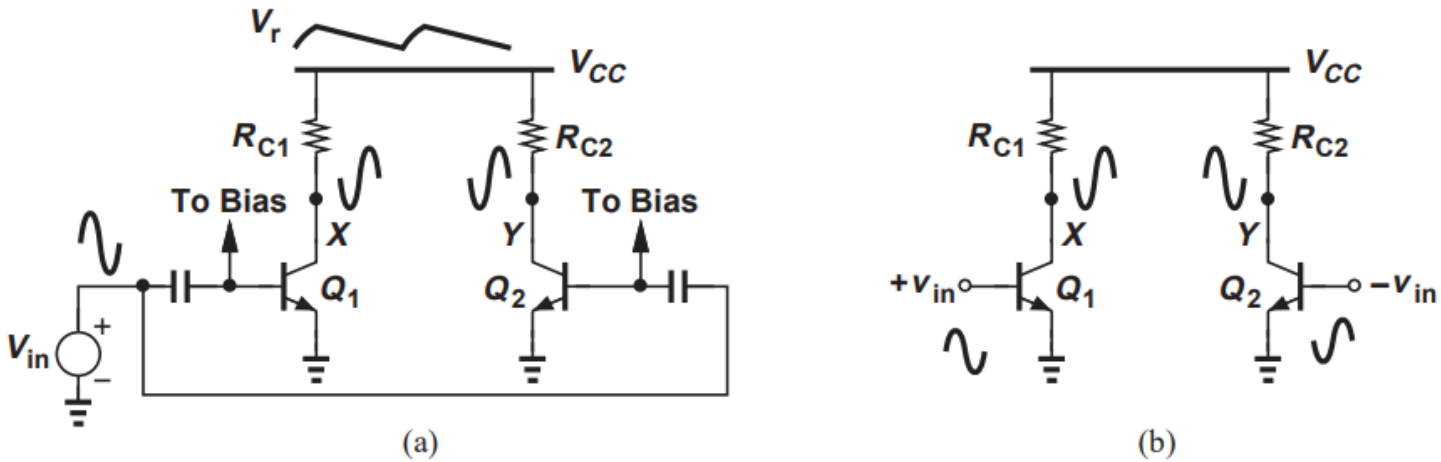


Figure 6: (a) Application of one input signal to two CE stages, (b) use of differential input signals. [4]

Differential Signals: An improvement of the solution

Unfortunately, the signal components at X and Y are in phase, canceling each other as they appear in $v_X - v_Y$:

$$\begin{cases} v_X = A_v \cdot v_{in} + v_r \\ v_Y = A_v \cdot v_{in} + v_r \end{cases} \tag{4}$$

For the signal components to enhance each other at the output, we can invert one of the input phases as shown in Fig. (6.b), obtaining:

$$\begin{cases} v_X = A_v \cdot v_{in} + v_r \\ v_Y = -A_v \cdot v_{in} + v_r \end{cases} \tag{5}$$

and hence

$$v_X - v_Y = 2 \cdot A_v v_{in} \tag{6}$$

Compared to the circuit of Fig. (5.a), this topology provides twice the output swing by exploiting the amplification capability of the duplicate stage.

Differential Signals: Waveform Analysis

Our thought process has led us to the specific waveforms in Fig. (6.b): the circuit senses 2 inputs that vary by equal and opposite amounts and generates 2 outputs that behave in a similar fashion.

These waveforms are examples of **"differential"** signals and stand in contrast to **"single-ended"** signals —the type to which we are accustomed from basic circuits and previous classes of this course.

More specifically, a single-ended signal is one measured with respect to the common ground (Fig. (7.a)) and "carried by one line", whereas a differential signal is measured between two nodes that have equal and opposite swings (Fig. (7.b)) and is thus "carried by two lines".

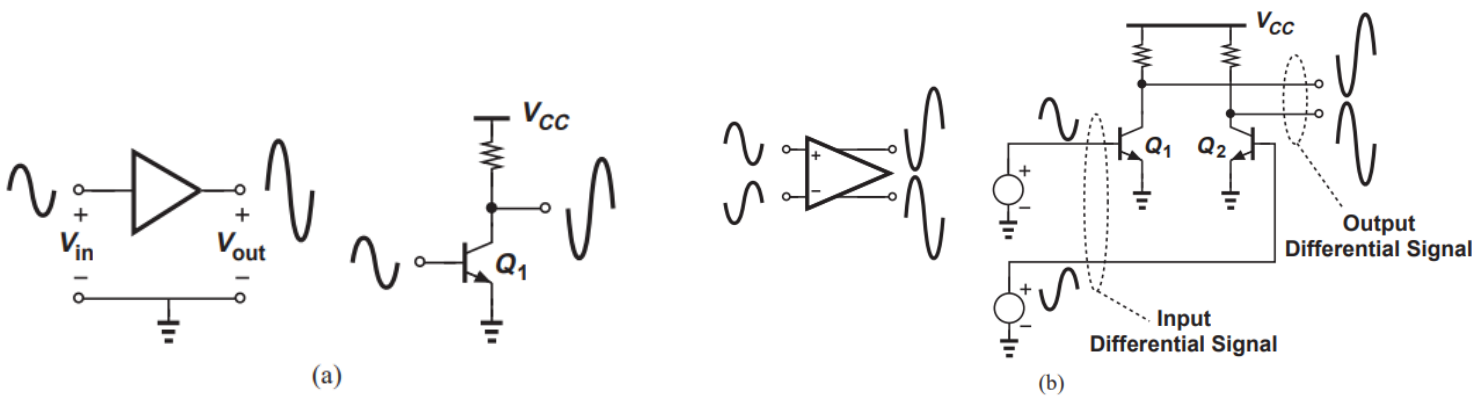


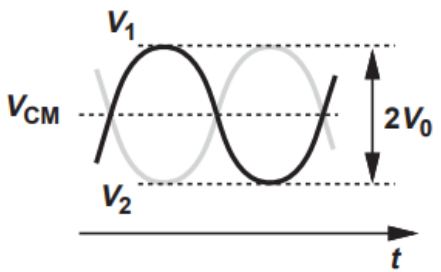
Figure 7: a) Single-ended signals, b) Differential signals. [5]

Differential Signals: Waveform Analysis

Here, V_1 and V_2 vary by equal and opposite amounts and have the same average (DC) level, V_{CM}, with respect to ground:

$$\begin{cases} V_1 = V_0 \cdot \sin(\omega t) + V_{CM} \\ V_2 = -V_0 \cdot \sin(\omega t) + V_{CM} \end{cases} \tag{7}$$

The DC voltage that is common to both V_1 and V_2 is called the **"common-mode (CM) level"**. That is, in the absence of differential signals, the 2 nodes remain at a potential equal to V_{CM} with respect to the global ground (see Fig. (8)).



(c)

Figure 8: Illustration of common-mode level. [5]

CMOS Differential Pair: Tail current source

Before formally introducing the differential pair, we must recognize that the circuit of Fig. (7.b) senses 2 inputs and can therefore serve as A_1 in Fig. (5.b). This observation leads to the differential pair.

While sensing and producing differential signals, the circuit of Fig. (7.b) suffers from some drawbacks. Fortunately, a simple modification yields an elegant, versatile topology: the sources of M_1 and M_2 are tied to a constant current source rather than to ground that we call I_{SS} - the "tail current source" as shown in Fig. (9).

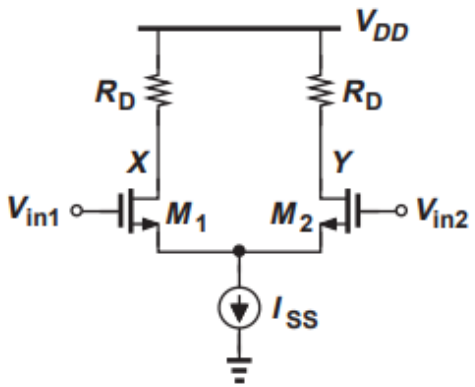


Figure 9: *MOS* Differential Amplifier. [1]

CMOS Differential Pair: Qualitative Analysis

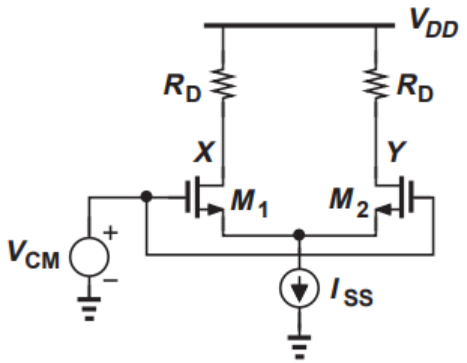


Figure 10: Response of MOS differential pair to input CM variation. [6]

Fig. (10) depicts the *MOS* pair with the two inputs tied to V_{CM}, where if we assume perfect symmetry, we have:

$$I_{D1} = I_{D2} = \frac{I_{SS}}{2} \tag{8}$$

and:

$$V_X = V_Y = V_{DD} - \frac{I_{SS}}{2} \cdot R_D \tag{9}$$

CMOS Differential Pair: Qualitative Analysis

Eq. (9) shows that a zero differential input gives a zero differential output. Note that the output common mode level is equal to: $V_{DD} - R_D \cdot I_{SS}/2$.

For our subsequent derivations, it is useful to compute the *"equilibrium overdrive voltage"* of M_1 and M_2, $(V_{GS} - V_{Th})_{equil}$. We assume $\lambda = 0$ and hence:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_{Th})^2 \tag{10}$$

Carrying a current of $I_{SS}/2$, each device exhibits an overdrive of:

$$(V_{GS} - V_{Th})_{equil} = \sqrt{\frac{I_{SS}}{\mu_n C_{ox}\left(\frac{W}{L}\right)}} \tag{11}$$

As expected, a greater tail current or a smaller W/L translates to a larger equilibrium overdrive.

CMOS Differential Pair: Qualitative Analysis

To guarantee that M_1 and M_2 operate in saturation, we require that their drain voltages not fall below: $V_{CM} - V_{Th}$:

$$V_{DS} = V_{DD} - R_D \cdot \frac{I_{SS}}{2} > V_{CM} - V_{Th} \tag{12}$$

It can also be observed that a change in V_{CM} cannot alter $I_{D1} = I_{D2} = I_{SS}/2$, leaving V_X and V_Y undisturbed. The circuit thus rejects input CM variations.

and what about the Differential Mode?

CMOS Differential Pair: Qualitative Analysis

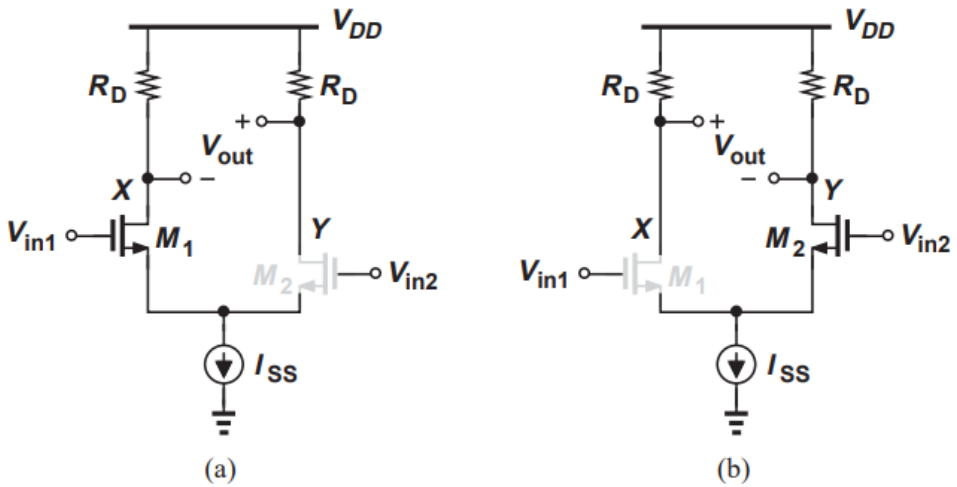


Figure 11: (a) Response of *MOS* differential pair to very positive input, (b) response of *MOS* differential pair to very negative input. [7]

Fig. (11) illustrates the response of the *MOS* pair to large differential inputs. If V_{in1} is well above V_{in2} [Fig. (11.a)], then M_1 carries the entire tail current, generating:

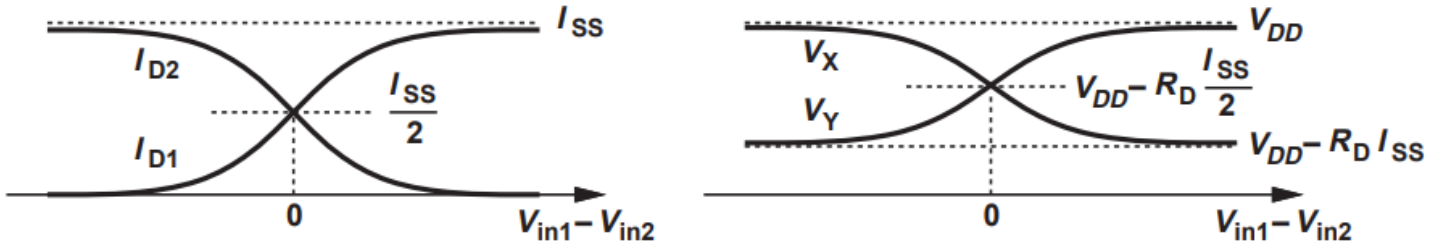
$$\begin{cases} V_X = V_{DD} - I_{SS} \cdot R_D \\ V_Y = V_{DD} \end{cases} \tag{13}$$

CMOS Differential Pair: Qualitative Analysis

Similarly, if V_{in2} is well above V_{in1} [Fig. (11.b)], then

$$\begin{cases} V_X = V_{DD} \\ V_Y = V_{DD} - I_{SS} \cdot R_D \end{cases} \tag{14}$$

The circuit therefore steers the tail current from one side to the other, producing a differential output in response to a differential input. Figure (12.c) sketches the characteristics of the circuit.



(c)

Figure 12: Qualitative plots of currents and voltages. [7]

CMOS Differential Pair: Qualitative Analysis

Let us now examine the circuit's behavior for a small input difference depicted in Fig. (13).

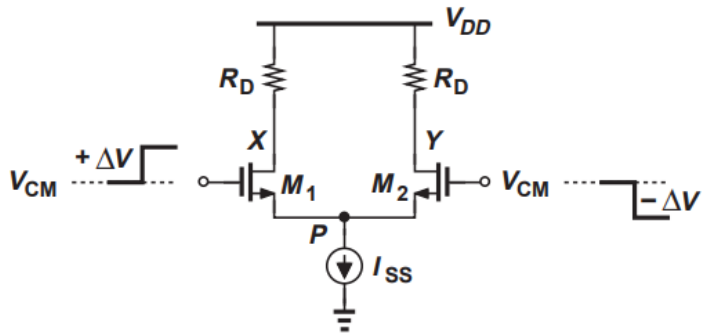


Figure 13: Qualitative plots of currents and voltages. [8]

In such scenario, the gate voltage of M_1 is raised from V_{CM} by ΔV while that of M_2 is lowered from V_{CM} by the same amount. We surmise that I_{D1} increases slightly and, since: $I_{D1} + I_{D2} = I_{SS}$, I_{D2} decreases by the same amount:

$$\begin{cases} I_{D1} = \frac{I_{SS}}{2} + \Delta I \\ I_{D2} = \frac{I_{SS}}{2} - \Delta I \end{cases} \tag{15}$$

CMOS Differential Pair: Qualitative Analysis

Now one may wonder how is ΔI related to ΔV? If the sources of M_1 and M_2 were directly tied to ground, then ΔI would simply be equal to: $g_m \Delta V$. In the differential pair, however, node P is free to rise or fall. We must therefore compute the change in V_P.

Suppose V_P rises by ΔV_P. As a result, the net increase in V_{GS1} is equal to $\Delta V - \Delta V_P$ and hence:

$$\Delta I_{D1} = g_m(\Delta V - \Delta V_P) \tag{16}$$

Similarly, the net decrease in V_{GS2} is equal to $\Delta V + \Delta V_P$, yielding:

$$\Delta I_{D2} = -g_m(\Delta V + \Delta V_P) \tag{17}$$

But recall from Eq. (15) that ΔI_{D1} must be equal to $-\Delta I_{D2}$, dictating that:

$$g_m(\Delta V - \Delta V_P) = -g_m(\Delta V + \Delta V_P) \rightarrow \Delta V_P = 0 \tag{18}$$

CMOS Differential Pair: Qualitative Analysis

Therefore, it follows that:

$$\begin{cases} \Delta I_{D1} = g_m \Delta V \\ \Delta I_{D2} = -g_m \Delta V \end{cases} \tag{19}$$

and:

$$\begin{cases} \Delta V_X = g_m \Delta V R_D \\ \Delta V_Y = -g_m \Delta V R_D \end{cases} \tag{20}$$

The differential output therefore goes from 0 to:

$$\Delta V_X - \Delta V_Y = -2 g_m R_D \Delta V \tag{21}$$

We define the small-signal differential gain of the circuit as:

$$A_v = \frac{\text{Change in Differential Output}}{\text{Change in Differential Input}} = \frac{-2 g_m R_D \Delta V}{2 \Delta V} = -g_m R_D \tag{22}$$

This expression is similar to that of the *common-source stage*.

CMOS Differential Pair: Large-Signal Analysis

Our objective here is to derive the input/output characteristics of the *MOS* pair as the differential input varies from very negative to very positive values. From Fig. (14), we have:

$$V_{out} = V_{out1} - V_{out2} = -R_D(I_{D1} - I_{D2}) \tag{23}$$

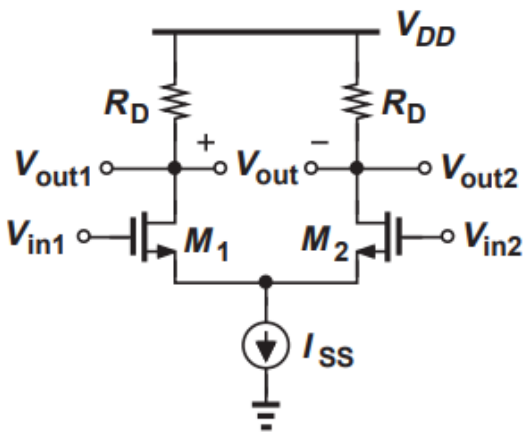


Figure 14: MOS differential pair for large-signal analysis. [9]

CMOS Differential Pair: Large-Signal Analysis

To obtain $I_{D1} - I_{D2}$, we neglect channel-length modulation and write a *KVL* around the input network and a *KCL* at the tail node:

$$V_{in1} - V_{GS1} = V_{in2} - V_{GS2} \tag{24}$$

$$I_{D1} + I_{D2} = I_{SS} \tag{25}$$

Since $I_D = (1/2)\mu_n C_{ox}(W/L)(V_{GS} - V_{Th})^2$, we have:

$$V_{GS} = V_{Th} + \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}} \tag{26}$$

Substituting for V_{GS1} and V_{GS2} in Eq. (24), we have:

$$V_{in1} - V_{in2} = V_{GS1} - V_{GS2} \tag{27}$$

$$V_{in1} - V_{in2} = \sqrt{\frac{2}{\mu_n C_{ox}\frac{W}{L}}}(\sqrt{I_{D1}} - \sqrt{I_{D2}}) \tag{28}$$

CMOS Differential Pair: Large-Signal Analysis

Squaring both sides yields:

$$(V_{in1} - V_{in2})^2 = \frac{2}{\mu_n C_{ox} \frac{W}{L}} (I_{D1} + I_{D2} - 2\sqrt{I_{D1} I_{D2}}) \tag{29}$$

Recognizing that $I_{D1} + I_{D2} = I_{SS}$, we obtain:

$$(V_{in1} - V_{in2})^2 = \frac{2}{\mu_n C_{ox} \frac{W}{L}} (I_{SS} - 2\sqrt{I_{D1} I_{D2}}) \tag{30}$$

We now find $\sqrt{I_{D1} I_{D2}}$,

$$4\sqrt{I_{D1} I_{D2}} = 2I_{SS} - \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 \tag{31}$$

square the result again,

$$16 I_{D1} I_{D2} = \left[2I_{SS} - \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 \right]^2 \tag{32}$$

CMOS Differential Pair: Large-Signal Analysis

and substitute $I_{SS} - I_{D1}$ for I_{D2},

$$16 I_{D1}(I_{SS} - I_{D1}) = \left[2I_{SS} - \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 \right]^2 \tag{33}$$

It follows that:

$$16 I_{D1}^2 - 16 I_{SS} I_{D1} + \left[2I_{SS} - \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 \right]^2 = 0 \tag{34}$$

and solving the quadratic expression, we have:

$$I_{D1} = \frac{I_{SS}}{2} \pm \frac{1}{4} \sqrt{4 I_{SS}^2 - \left[\mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 - 2I_{SS} \right]^2} \tag{35}$$

Note that the only solution that is valid, is the one with the sum of the two terms:

$$I_{D1} = \frac{I_{SS}}{2} + \frac{V_{in1} - V_{in2}}{4} \sqrt{\mu_n C_{ox} \frac{W}{L} \left[4I_{SS} - \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in2})^2 \right]} \tag{36}$$

CMOS Differential Pair: Large-Signal Analysis

The symmetry of the circuit also implies that:

$$I_{D2} = \frac{I_{SS}}{2} + \frac{V_{in2} - V_{in1}}{4}\sqrt{\mu_n C_{ox}\frac{W}{L}\left[4I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in2} - V_{in1})^2\right]} \tag{37}$$

That is,

$$I_{D1} - I_{D2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})\sqrt{\frac{4I_{SS}}{\mu_n C_{ox}\frac{W}{L}} - (V_{in1} - V_{in2})^2} \tag{38}$$

Where Eqs. (36) - (38) form the foundation of our understanding of the *MOS* differential pair

CMOS Differential Pair: Large-Signal Analysis

Let us now examine Eq. (38) closely.

- As expected from the characteristics in Fig. (15.c), the right-hand side is an odd (symmetric) function of $V_{in1} - V_{in2}$, dropping to zero for a zero input difference.

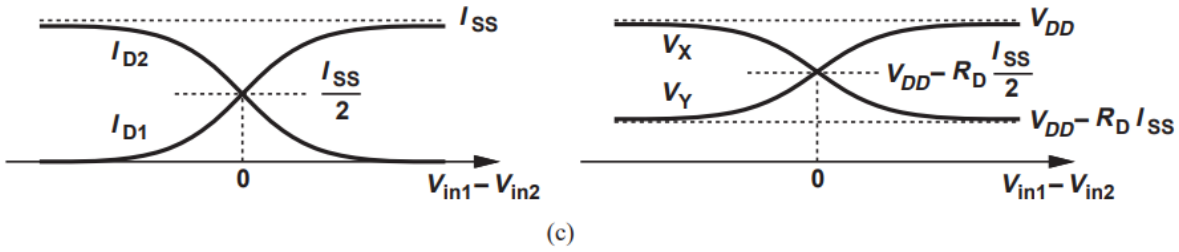


Figure 15: Qualitative plots of currents and voltages. [7]

- But, *can the difference under the square root vanish, too?* That would suggest that: $I_{D1} - I_{D2}$ falls to zero as $(V_{in1} - V_{in2})^2$ reaches $4I_{SS}/(\mu_n C_{ox} W/L)$, an effect not predicted by our qualitative sketches in Fig. (15.c). Furthermore, it appears that the argument of the square root becomes negative as $(V_{in1} - V_{in2})^2$ exceeds this value! $\rightarrow$ **How should these results be interpreted?**

CMOS Differential Pair: Large-Signal Analysis

- Implicit in our foregoing derivations is the assumption both transistors are on. However, as $|V_{in1} - V_{in2}|$ rises, at some point M_1 or M_2 turns off, violating the above equations.
- We must therefore determine the input difference that places one of the transistors at the edge of conduction.
  - This can be accomplished by equating Eqs. (36) and (37), or (38) to I_{SS}, but this leads to lengthy algebra...
  - Instead, we recognize from Fig. (16) that if, for example, M_1 approaches the edge of conduction, then its gate-source voltage falls to a value equal to V_{Th}.

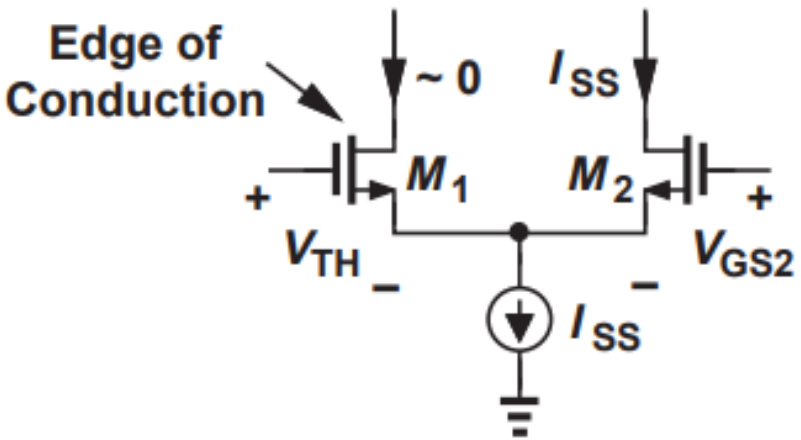


Figure 16: MOS differential pair with one device off. [10]

CMOS Differential Pair: Large-Signal Analysis

Also, the gate-source voltage of M_2 must be sufficiently large to accommodate a drain current of I_{SS}:

$$\begin{cases} V_{GS1} = V_{Th} \\ V_{GS2} = V_{Th} + \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} \end{cases} \tag{39}$$

It follows from Eq. (24) that:

$$|V_{in1} - V_{in2}|_{max} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} \tag{40}$$

- where $|V_{in1} - V_{in2}|_{max}$ denotes the input difference that places one transistor at the edge of conduction.
- Eq. (40) is invalid for input differences greater than this value.
- Indeed, substituting from Eq. (40) in (38) also yields $|I_{D1} - I_{D2}| = I_{SS}$. We also note that $|V_{in1} - V_{in2}|_{max}$ can be related to the equilibrium overdrive [Eq. (11)] as follows:

$$|V_{in1} - V_{in2}|_{max} = \sqrt{2}(V_{GS} - V_{Th})_{equil.} \tag{41}$$

CMOS Differential Pair: Large-Signal Analysis

The above findings are very important and stand in contrast to the behavior of the bipolar differential pair:

- The *MOS* pair steers all of the tail current for $|V_{in1} - V_{in2}|_{max}$ whereas the bipolar counterpart only approaches this condition for a finite input difference.

$$|V_{in1} - V_{in2}|_{max} = \sqrt{2}(V_{GS} - V_{Th})_{equil}. \tag{42}$$

- Equation (42) provides a great deal of intuition into the operation of the *MOS* pair: Specifically, we plot I_{D1} and I_{D2} as in Fig. (17.a), where $\Delta V_{in} = V_{in1} - V_{in2}$, arriving at the differential characteristics in Figs. (17.b) and (17.c). The circuit thus behaves linearly for small values of ΔV_{in} and becomes completely nonlinear for $\Delta V_{in} > \Delta V_{in,max}$. In other words, $\Delta V_{in,max}$ serves as an absolute bound on the input signal levels that have any effect on the output.

CMOS Differential Pair: Large-Signal Analysis

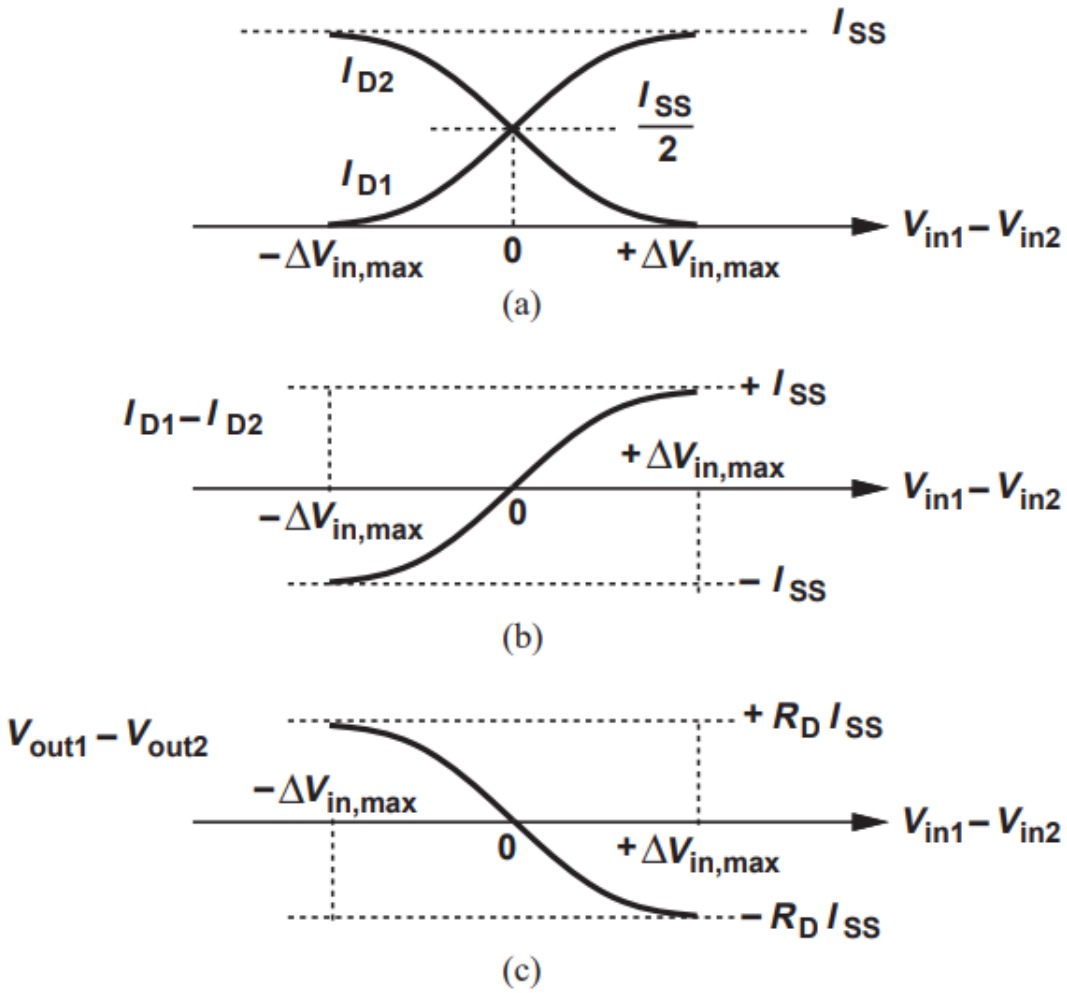


Figure 17: Variation of (a) drain currents, (b) the difference between drain currents, and (c) differential output voltage as a function of input. [11]

Conclusion

- Thank you for listening! If you have any questions about the topics in this presentation, feel free to ask and we can discuss them anytime.

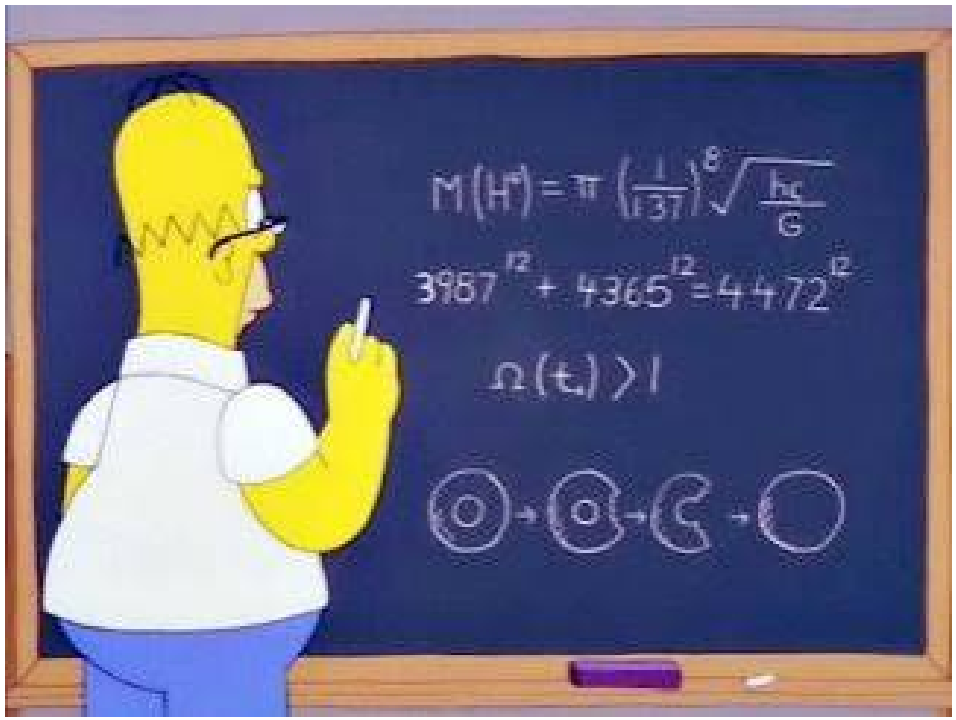


Figure 18: Me explaining *MOS* Differential Pairs.

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